MODULE 16, LESSON 1

AI 101 FOR MECHANICAL ENGINEERS

APPLICATIONS WITH DIFFERENT EVIDENCE NEEDS

Gemini Notebook

Engineering Implementation Hierarchy

1. **Surrogate Models**
   Approximating the physical system

2. **Optimization Fundamentals**
   Establishing search constraints

3. **Advanced Integrations**
   Connecting digital models to physical hardware

Gemini Notebook

01

SECTION 01

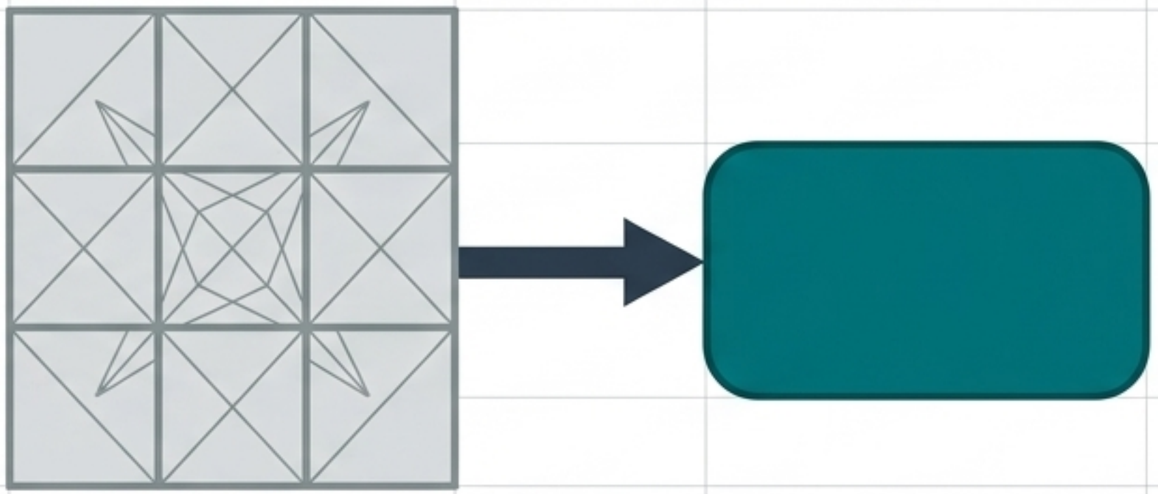
Surrogate Models

Approximating system physics to enable rapid computational search.

Gemini Notebook

Surrogate Model Definition

- A fast approximation to costly experiments or simulators.
- Enables cheap computational search across design parameter spaces.
- **Example:** Predict pressure drop ΔP [Pa] during geometry search.



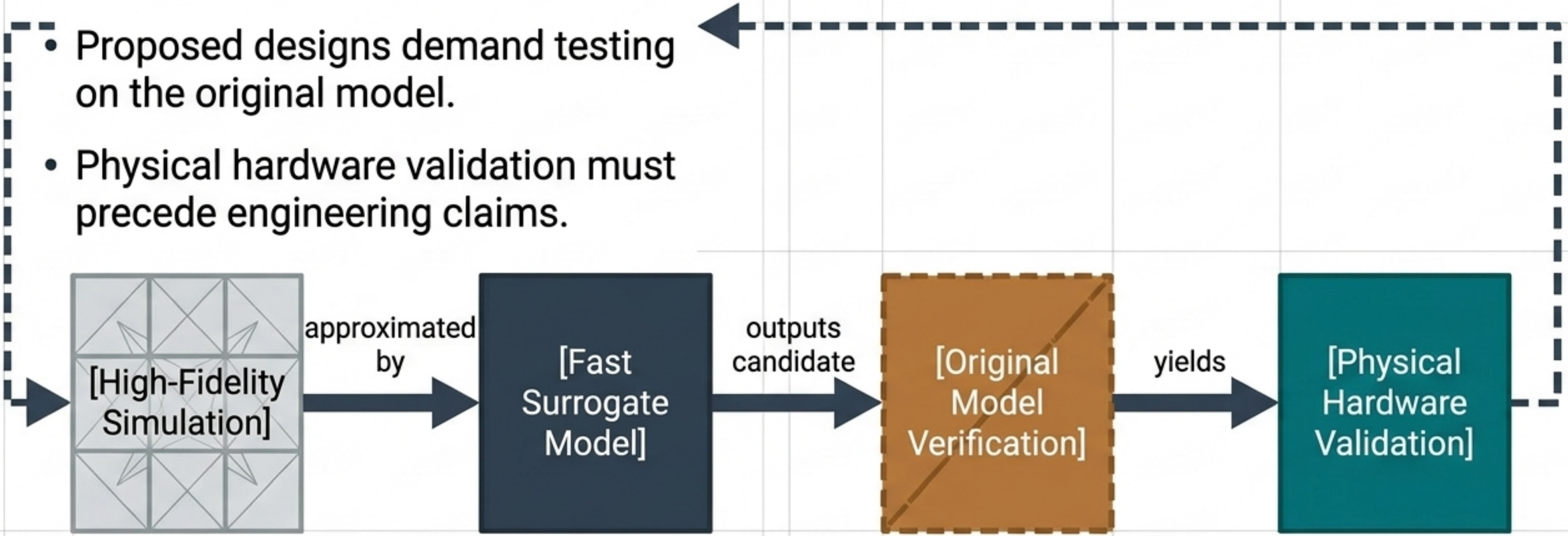
High-Fidelity Physics Simulator

Fast Surrogate Approximation

Gemini Notebook

The Surrogate Verification Loop

- Surrogates approximate physics but require rigorous verification.
- Proposed designs demand testing on the original model.
- Physical hardware validation must precede engineering claims.



Gemini Notebook

02

SECTION 02

Optimization Fundamentals

Defining objectives, constraints, and operational bounds.

Gemini Notebook

Objectives and Constraints

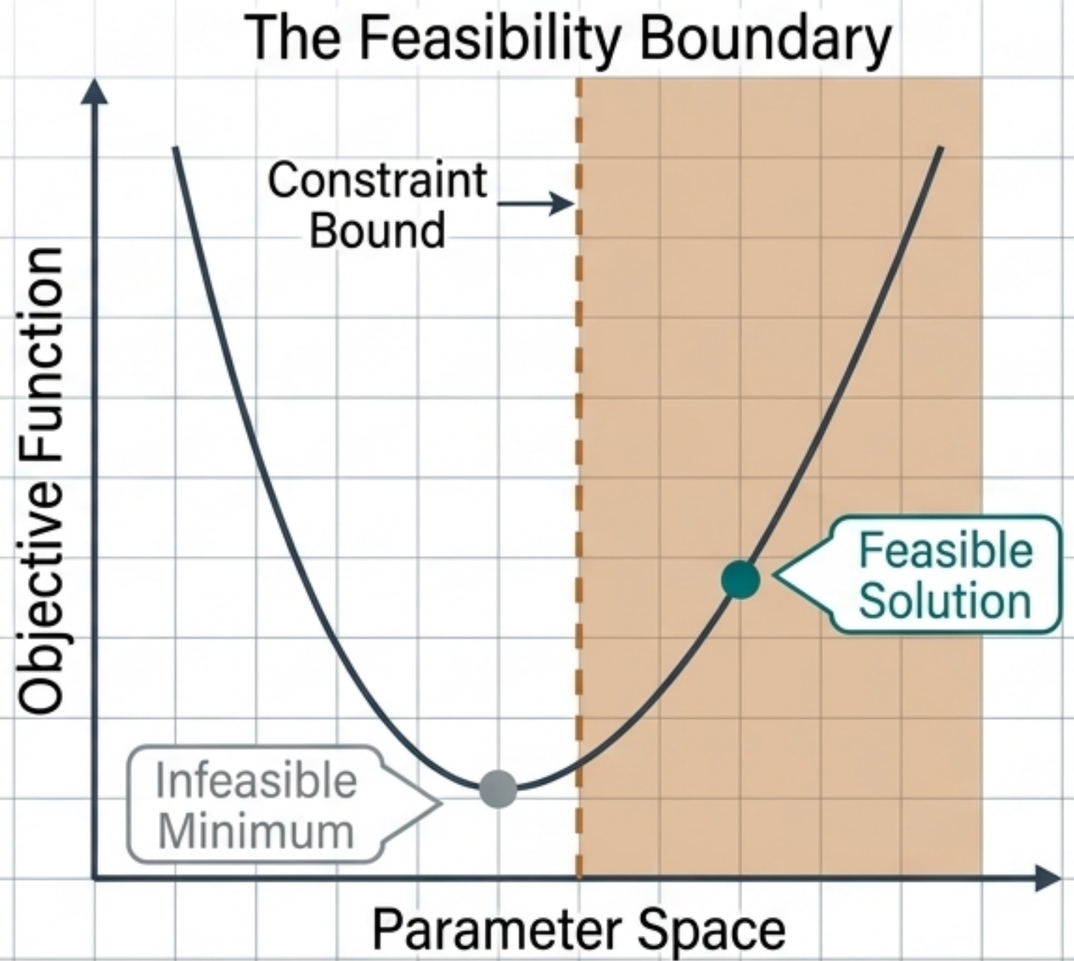
- **Optimization objective**: A targeted quantity to improve.
- Example: Minimize system energy consumption E [J].
- **Constraint:** A strict requirement limiting permitted choices.
- **Bounds** define the outer limits of permissible parameters.
- Example: Keep mechanical stress σ [Pa] below limit σ_{max}.

Parameter Type	Definition	Mathematical Example
Objective	Quantity to improve	Min E [J]
Constraint	Permitted-choice requirement	$\sigma < \sigma_{max}$ [Pa]

Gemini Notebook

Defining Feasibility

- Feasibility confirms satisfaction of stated bounds and constraints.
- Example: Verified stress strictly meets the σ_{max} limit.
- A low predicted objective is never inherent evidence of feasibility.



Gemini Notebook

03

SECTION 03

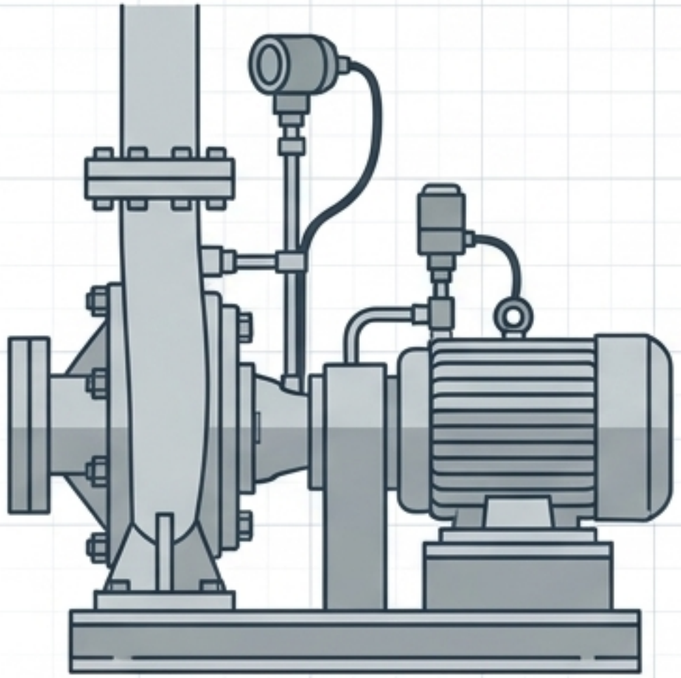
Advanced Integrations

Linking digital environments to physical counterparts.

Gemini Notebook

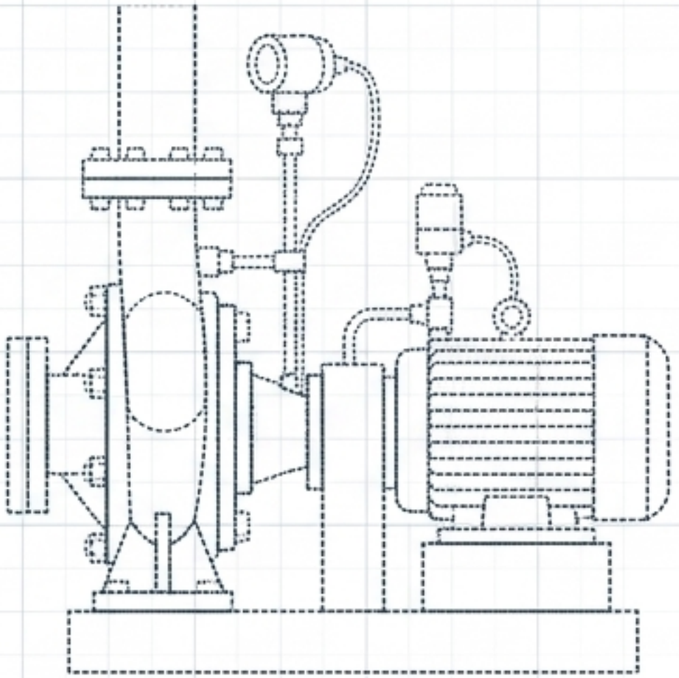
Digital Twins via Data Connections

- **Digital twin**: Digital representation connected to physical counterpart.
- Connection strictly relies on continuous, bidirectional data transfer.
- Isolated geometry representation cannot function as a twin.



[Physical System]

[Active Data Bridge]



[Digital Representation]

Gemini Notebook

System Comparison: Twin vs. CAD

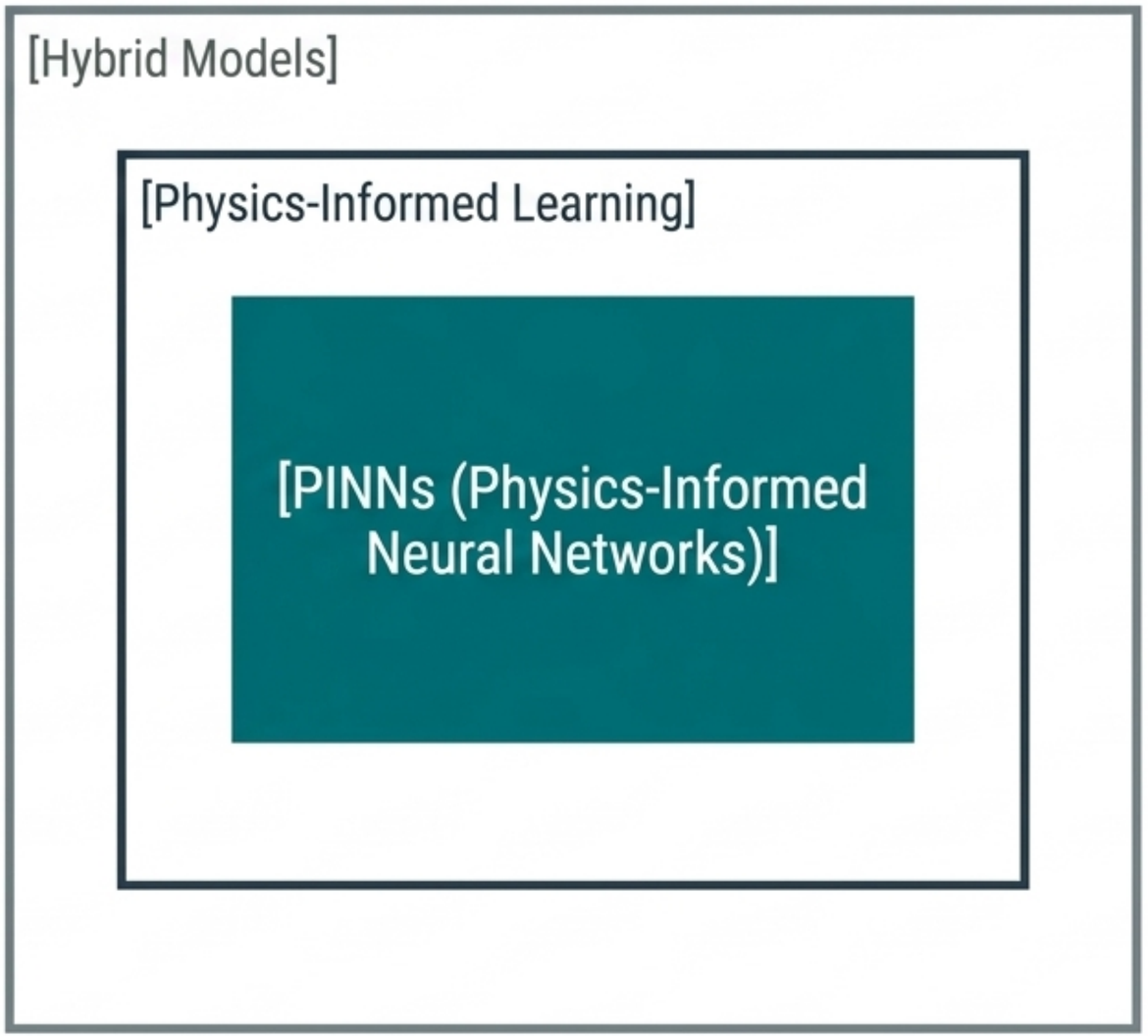
- Standalone CAD files lack required physical data integration.
- True twins require active physical counterpart links.

System Feature	Isolated CAD Model	Digital Twin
Primary Function	Shape representation	System monitoring
Data Connection Status	None (Isolated)	Active bidirectional
Physical Counterpart Link	Absent	Present

Gemini Notebook

Physics-Informed Learning Hierarchy

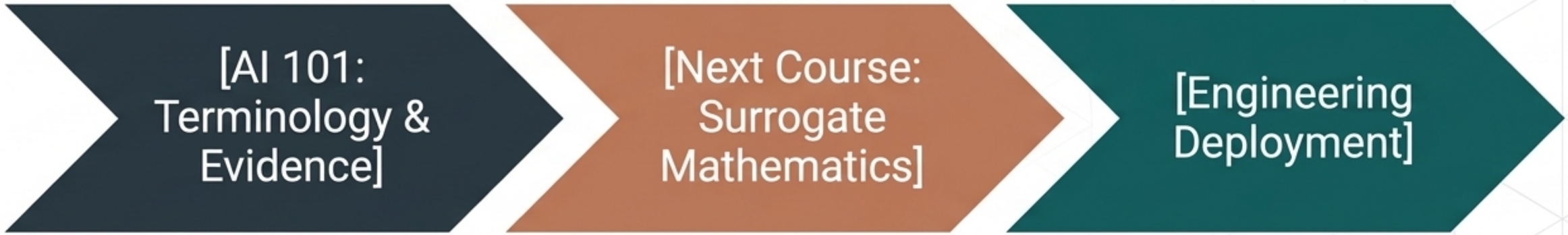
- Integrates physical equations, constraints, or structure into learning.
- Example: Penalizing a mass conservation residual during neural conservation residual during neural network training.
- Physics-Informed Neural Networks (PINNs) are one specific family.
- PINNs do not represent all hybrid learning models.



Gemini Notebook

Progression to Deeper Mathematics

- This module establishes core terminology and strict evidence habits.
- Hardware deployment requires extensive testing and engineering fundamentals.
- Subsequent courses demand rigorous surrogate modeling mathematics.
- Advanced Bayesian optimization requires strict mathematical foundations.

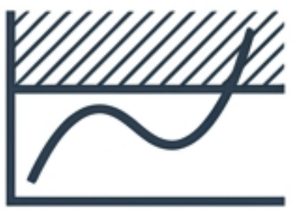


Gemini Notebook

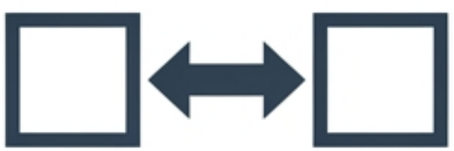
Key Points



1. Surrogates approximate expensive physics but demand rigorous verification loops.



2. Feasibility strictly requires satisfying constraints, independent of optimization objectives.



3. Digital twins necessitate active, bidirectional data bridges to physical counterparts.

Gemini Notebook